

CS 109ACE Week 2 Practice Problems

1 Axioms and Equally Likely Outcomes

Problem 1: Quick Practice

(slide 36)

Roll a fair six-sided die 4 times. What is the probability that you see at least one 6?

Solution. Let E be the event “at least one 6.” It is easier to compute the complement E^c = “no 6 in any of the 4 rolls.” On one roll, $P(\text{not } 6) = \frac{5}{6}$, and the rolls are independent, so

$$P(E^c) = \left(\frac{5}{6}\right)^4 = \frac{625}{1296}.$$

Therefore

$$P(E) = 1 - P(E^c) = 1 - \left(\frac{5}{6}\right)^4 = \frac{671}{1296} \approx 0.518.$$

■

Problem 2: Let’s do it together

(slide 40)

Roll two fair six-sided dice. Which outcomes make up the event “the sum is 7”? What is its probability?

Solution. The sample space is all ordered pairs, $S = \{(i, j) : 1 \leq i, j \leq 6\}$, so $|S| = 36$, and all outcomes are equally likely. The event is

$$E = \{(1, 6), (2, 5), (3, 4), (4, 3), (5, 2), (6, 1)\}, \quad |E| = 6,$$

so

$$P(E) = \frac{|E|}{|S|} = \frac{6}{36} = \frac{1}{6}.$$

Trap: the 11 possible *sums* $2, \dots, 12$ are not equally likely (e.g. $P(\text{sum} = 2) = \frac{1}{36}$), so $P(\text{sum} = 7) \neq \frac{1}{11}$. ■

2 Review Problems

Problem 3: Frisbee

(slide 74)

You flip two frisbees. Each frisbee independently lands “heads” with probability 0.6 (and “tails” with probability 0.4). The result is *even* if both frisbees land heads or both land tails. What is $P(\text{even})$?

Solution. The event “even” is the union of two mutually exclusive events: both heads (HH) and both tails (TT). Since the flips are independent,

$$P(HH) = 0.6 \cdot 0.6 = 0.36, \quad P(TT) = 0.4 \cdot 0.4 = 0.16.$$

Because HH and TT are mutually exclusive, we add:

$$P(\text{even}) = P(HH) + P(TT) = 0.36 + 0.16 = 0.52.$$

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Problem 4: Network Reliability

(slide 76)

A network connects A to B through n routers arranged in *parallel*. Router i works independently with probability p_i , for $1 \leq i \leq n$. Let E be the event that there is a functional path from A to B . What is $P(E)$?

Solution. Let F_i be the event that router i works. Because the routers are in parallel, a path exists if at least one router works: $E = F_1 \cup F_2 \cup \dots \cup F_n$. Unions of independent events are awkward, so use the complement and De Morgan's law:

$$E^c = F_1^c \cap F_2^c \cap \dots \cap F_n^c.$$

By independence,

$$P(E^c) = \prod_{i=1}^n P(F_i^c) = \prod_{i=1}^n (1 - p_i), \quad \text{so} \quad P(E) = 1 - \prod_{i=1}^n (1 - p_i).$$

Check: with 3 routers each working with probability 0.9, $P(E) = 1 - (0.1)^3 = 0.999$.

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Problem 5: Zika Testing

(slide 78)

A patient tests positive for Zika. Suppose 0.8% of people have Zika. The test is positive 90% of the time for people who have Zika, and 7% of the time for people who do not (false positives). What is the probability that the patient has Zika, given the positive test?

Solution. Let Z = “has Zika” and T = “tests positive.” We are given $P(Z) = 0.008$, $P(T | Z) = 0.90$, and $P(T | Z^c) = 0.07$. By the Law of Total Probability,

$$P(T) = P(T | Z)P(Z) + P(T | Z^c)P(Z^c) = 0.9(0.008) + 0.07(0.992) = 0.0072 + 0.06944 = 0.07664.$$

By Bayes' Theorem,

$$P(Z | T) = \frac{P(T | Z) P(Z)}{P(T)} = \frac{0.0072}{0.07664} \approx 0.094 \approx 9\%.$$

Intuition: out of 1000 people, about 7.2 are true positives but about 69.4 are false positives, so most positive tests come from people without Zika.

■

3 More Practice

3.1 Warm-ups: Axioms and Counting

Problem 6: Warm-up 1: 8-Sided Die

(slide 81)

Roll a fair 8-sided die with faces $1, 2, \dots, 8$. Let A be the event that the result is even, and B the event that the result is at least 6. Find:

- (a) $P(A)$
- (b) $P(B)$
- (c) $P(A \cap B)$
- (d) $P(A \cup B)$

Hint: list the outcomes in each event.

Solution. $S = \{1, \dots, 8\}$ with equally likely outcomes, $A = \{2, 4, 6, 8\}$, $B = \{6, 7, 8\}$.

- (a) $P(A) = \frac{4}{8} = \frac{1}{2}$.
- (b) $P(B) = \frac{3}{8}$.
- (c) $A \cap B = \{6, 8\}$, so $P(A \cap B) = \frac{2}{8} = \frac{1}{4}$.
- (d) By inclusion-exclusion, $P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{4}{8} + \frac{3}{8} - \frac{2}{8} = \frac{5}{8}$. Check: $A \cup B = \{2, 4, 6, 7, 8\}$ has 5 outcomes.

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Problem 7: Warm-up 2: Inviting Friends

(slide 83)

You have 6 friends: George, Hannah, Kevin, Shree, Ulo, and Sam.

- (a) How many ways can you choose 2 of them to invite?
- (b) Shree and Sam will only come together (both or neither). How many ways can you choose 2 friends to invite?

Solution.

- (a) Order does not matter, so the count is $\binom{6}{2} = 15$.
- (b) Split into cases. Either you invite Shree and Sam together (1 way), or you invite neither and choose 2 of the other 4 ($\binom{4}{2} = 6$ ways). Total: $1 + 6 = 7$.

■

Problem 8: Warm-up 3: Entry Codes

(slide 85)

An entry code is a string of 6 digits (0–9); codes may start with 0.

- (a) How many codes are there in total?
- (b) How many codes consist of the same two digits repeated three times (e.g. 121212 or 333333)?
- (c) How many codes use each of the digits 1, 2, 3, 4, 5, 6 exactly once?
- (d) *Stretch:* a code is *valid* if its digits sum to 9. If a code is chosen uniformly at random, what is the probability that it is valid?

Solution.

- (a) Each of the 6 positions has 10 choices: $10^6 = 1,000,000$.
- (b) The code has the form $ababab$, determined by the pair (a, b) : $10 \cdot 10 = 100$ codes. (333333 corresponds only to $a = b = 3$, so it is counted exactly once.)
- (c) The code is an ordering of the 6 digits: $6! = 720$.

- (d) We need the number of solutions to $d_1 + \cdots + d_6 = 9$ with $d_i \geq 0$ integers. Since the total is 9, no digit can exceed 9, so the constraint $d_i \leq 9$ is automatic. By stars and bars, the count is $\binom{9+5}{5} = \binom{14}{5} = 2002$. Hence

$$P(\text{valid}) = \frac{2002}{10^6} \approx 0.002.$$

■

Problem 9: Six-Card Poker

(slide 89)

You are dealt an unordered 6-card hand from a standard 52-card deck.

- (a) What is the probability of *three pair* (2 cards each of three different ranks)?
 (b) What is the probability of a *pyramid* (3 cards of one rank, 2 of a second rank, and 1 of a third rank)?

Solution. All hands are equally likely, with $|S| = \binom{52}{6} = 20,358,520$.

- (a) Choose the 3 ranks (unordered, since they play the same role), then 2 of 4 suits for each:

$$|E| = \binom{13}{3} \binom{4}{2}^3 = 286 \cdot 216 = 61,776, \quad P = \frac{61,776}{20,358,520} = \frac{594}{195,755} \approx 0.0030.$$

- (b) The three ranks play different roles (triple, pair, single), so choose them in order, then the suits:

$$|E| = 13 \cdot 12 \cdot 11 \cdot \binom{4}{3} \binom{4}{2} \binom{4}{1} = 1716 \cdot 96 = 164,736,$$

$$P = \frac{164,736}{20,358,520} = \frac{1584}{195,755} \approx 0.0081.$$

■

Problem 10: Warm-up 4: Straights

(slide 91)

You are dealt a 5-card hand from a standard 52-card deck. A *straight* is 5 cards of consecutive ranks; the ace may be low ($A, 2, 3, 4, 5$) or high ($10, J, Q, K, A$), and suits do not matter (so straight flushes count). What is $P(\text{straight})$? Describe the sample space S and the event E first.

Solution. S is the set of unordered 5-card hands, $|S| = \binom{52}{5} = 2,598,960$, all equally likely. E is the set of straights. A straight is determined by its lowest rank, which can be $A, 2, \dots, 10$: 10 choices. Each of the 5 cards can then be any of 4 suits: 4^5 choices. So

$$|E| = 10 \cdot 4^5 = 10,240, \quad P(E) = \frac{10,240}{2,598,960} \approx 0.0039.$$

There is no extra factor of 2: naming each straight by its lowest card counts every hand exactly once.

■

Problem 11: Warm-up 5: Casino

(slide 93)

You place 2 tokens on the possible sums of two dice, $2, \dots, 12$ (both tokens may go on the same number). The dealer rolls two dice twice. Each time the sum of a roll has a token on it, one token is removed from that number and you win \$1000.

- (a) With the strategy (7, 7), find $P(\$2000)$, $P(\$0)$, and $P(\$1000)$.
- (b) Repeat for the strategy (2, 7).
- (c) *Preview of expectation:* which strategy wins more on average?

Solution. For one roll, $P(\text{sum} = 7) = \frac{6}{36}$ and $P(\text{sum} = 2) = \frac{1}{36}$; the two rolls are independent.

- (a) Both tokens are on 7, so every 7 pays. $P(\$2000) = \left(\frac{6}{36}\right)^2 = \frac{1}{36}$, $P(\$0) = \left(\frac{30}{36}\right)^2 = \frac{25}{36}$, and $P(\$1000) = 1 - \frac{1}{36} - \frac{25}{36} = \frac{10}{36}$.
- (b) To win \$2000 you need one roll to be 2 and the other to be 7, in either order: $P(\$2000) = 2 \cdot \frac{1}{36} \cdot \frac{6}{36} = \frac{12}{1296} = \frac{1}{108}$. You win \$0 if neither roll is 2 or 7: $P(\$0) = \left(\frac{29}{36}\right)^2 = \frac{841}{1296}$. So $P(\$1000) = 1 - \frac{12}{1296} - \frac{841}{1296} = \frac{443}{1296}$.
- (c) Average winnings:

$$(7, 7) : 2000 \cdot \frac{1}{36} + 1000 \cdot \frac{10}{36} \approx \$333.33, \quad (2, 7) : 2000 \cdot \frac{12}{1296} + 1000 \cdot \frac{443}{1296} \approx \$360.34.$$

So (2, 7) wins more on average: stacking both tokens on 7 wastes chances. The best placement is (6, 7), at about \$564. ■

3.2 Conditional Probability

Problem 12: Conditional 1: Three Coin Flips

(slide 96)

A fair coin is flipped 3 times. Given that at least one flip is heads, what is the probability that at least two flips are heads?

Solution. Let F = “at least one head” and E = “at least two heads.” All 8 outcomes are equally likely, and F excludes only TTT , so $|F| = 7$. $E \cap F = E = \{HHT, HTH, THH, HHH\}$, so

$$P(E | F) = \frac{|E \cap F|}{|F|} = \frac{4}{7} \approx 0.571.$$

Compare with the unconditional $P(E) = \frac{4}{8} = \frac{1}{2}$: conditioning on F raises the probability. ■

Problem 13: Conditional 2: Two Dice

(slide 98)

Roll two fair dice. Let X be the first die and Y the second. Find:

- (a) $P(X > 3 | Y = 2)$
- (b) $P(X = 4 | X + Y = 8)$

Solution.

- (a) The dice are independent, so knowing Y tells us nothing about X : $P(X > 3 | Y = 2) = P(X > 3) = \frac{3}{6} = \frac{1}{2}$.
- (b) $\{X + Y = 8\} = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$ has 5 equally likely outcomes, and exactly one of them has $X = 4$. So

$$P(X = 4 | X + Y = 8) = \frac{P(X = 4, X + Y = 8)}{P(X + Y = 8)} = \frac{1/36}{5/36} = \frac{1}{5}.$$

**Problem 14: Conditional 3: Back to the 8-Sided Die***(slide 100)*

Use the 8-sided die from Warm-up 1: A = even, B = at least 6, with $P(A) = \frac{1}{2}$, $P(B) = \frac{3}{8}$, $P(A \cap B) = \frac{1}{4}$.

- (a) Find $P(A | B)$.
- (b) Find $P(B | A)$.
- (c) Is $P(A \cap B) = P(A)P(B)$ always, never, or sometimes true?
- (d) Is $P(A \cap B) = P(A | B)P(B)$ always, never, or sometimes true?
- (e) Let C be the event that the result is a perfect cube (1 or 8). Find $P(C | A \cap B)$.

Solution.

- (a) $P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{1/4}{3/8} = \frac{2}{3}$.
- (b) $P(B | A) = \frac{P(A \cap B)}{P(A)} = \frac{1/4}{1/2} = \frac{1}{2}$.
- (c) *Sometimes:* it holds exactly when A and B are independent. Here $P(A)P(B) = \frac{1}{2} \cdot \frac{3}{8} = \frac{3}{16} \neq \frac{1}{4}$, so A and B are dependent.
- (d) *Always:* this is the definition of conditional probability rearranged (the chain rule), valid whenever $P(B) > 0$.
- (e) $A \cap B = \{6, 8\}$, and only 8 is a cube, so $P(C | A \cap B) = \frac{1}{2}$.

**3.3 Chain Rule****Problem 15: Chain Rule 1: The Late Cat***(slide 102)*

It rains (R) with probability $\frac{1}{3}$. If it rains, there is traffic (T) with probability $\frac{1}{2}$; if not, with probability $\frac{1}{4}$. The cat is late (L) with probability $\frac{1}{2}$ if there is rain and traffic, $\frac{1}{8}$ if there is neither rain nor traffic, and $\frac{1}{4}$ otherwise.

- (a) What is the probability that it does not rain, there is traffic, and the cat is not late?

Solution.

- (a) By the chain rule,

$$P(R^c \cap T \cap L^c) = P(R^c) P(T | R^c) P(L^c | R^c \cap T) = \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{3}{4} = \frac{1}{8}.$$

**Problem 16: Chain Rule 2: White Elephant***(slide 104)*

n people each bring a gift to a pile. In turn, each person is assigned a gift uniformly at random from those remaining in the pile (possibly their own).

- (a) What is the probability that everyone gets their own gift?
- (b) What is the probability that exactly $n - 1$ people get their own gift?
- (c) What is the probability that exactly $n - 2$ people get their own gift?
- (d) If you can choose your turn, which turn maximizes your chance of getting your own gift?

Solution. The process produces a uniformly random assignment (permutation) of gifts to people.

- (a) By the chain rule, the first person gets theirs with probability $\frac{1}{n}$, then the second with probability $\frac{1}{n-1}$ given that, and so on: $\frac{1}{n} \cdot \frac{1}{n-1} \cdots \frac{1}{1} = \frac{1}{n!}$.
- (b) 0. If $n - 1$ people have their own gifts, the only remaining gift belongs to the last person.
- (c) Exactly two people must swap gifts and everyone else gets their own. There are $\binom{n}{2}$ ways to choose the pair, and each such assignment has probability $\frac{1}{n!}$:

$$\frac{\binom{n}{2}}{n!} = \frac{1}{2(n-2)!}.$$

- (d) It does not matter. If you go k -th, your gift is still in the pile with probability $\frac{n-k+1}{n}$, and then you draw it with probability $\frac{1}{n-k+1}$:

$$P(\text{own gift}) = \frac{n-k+1}{n} \cdot \frac{1}{n-k+1} = \frac{1}{n} \quad \text{for every } k.$$

Bonus: $P(\text{nobody gets their own gift})$ is $\frac{1}{3}$ for $n = 3$, $\frac{3}{8}$ for $n = 4$, and tends to $\frac{1}{e} \approx 0.368$ as $n \rightarrow \infty$. ■

3.4 Independence

Problem 17: Independence 1: Penny and Nickel

(slide 107)

Flip a fair penny and a fair nickel. Let A = the penny lands tails, B = the nickel lands tails, C = neither lands tails, and D = exactly one lands tails. Which pairs of events are independent?

Solution. $S = \{HH, HT, TH, TT\}$ (penny first), all equally likely. $A = \{TH, TT\}$, $B = \{HT, TT\}$, $C = \{HH\}$, $D = \{HT, TH\}$, so $P(A) = P(B) = P(D) = \frac{1}{2}$ and $P(C) = \frac{1}{4}$.

- A, B : $P(A \cap B) = P(\{TT\}) = \frac{1}{4} = P(A)P(B)$. Independent.
- A, D : $P(A \cap D) = P(\{TH\}) = \frac{1}{4} = P(A)P(D)$. Independent.
- B, D : $P(B \cap D) = P(\{HT\}) = \frac{1}{4} = P(B)P(D)$. Independent.
- C with A, B , or D : the events are disjoint, so the intersection has probability $0 \neq P(C) \cdot \frac{1}{2}$. Dependent.

So $\{A, B\}$, $\{A, D\}$, $\{B, D\}$ are independent; every pair involving C is dependent. (Disjoint events with positive probability are never independent.) ■

Problem 18: Independence 2: mRNA

(slide 109)

An mRNA strand of 8 letters is generated, each letter independently, with $P(A) = P(U) = \frac{1}{3}$ and $P(C) = P(G) = \frac{1}{6}$. Let V_i be the event that the sequence **AGUA** starts at position i .

- (a) Find $P(V_1)$.
- (b) Are V_1 and V_2 independent?
- (c) Are V_1 and V_4 independent?
- (d) Are V_1 and V_5 independent?

Solution.

- (a) $P(V_1) = P(\mathbf{A})P(\mathbf{G})P(\mathbf{U})P(\mathbf{A}) = \frac{1}{3} \cdot \frac{1}{6} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{162}$. By the same computation, $P(V_i) = \frac{1}{162}$ for each $i \leq 5$.
- (b) Dependent. If V_1 occurs, position 2 is $\mathbf{G}$, but V_2 needs position 2 to be $\mathbf{A}$. So $P(V_2 | V_1) = 0 \neq P(V_2)$.
- (c) Dependent. The events overlap at position 4, where both need $\mathbf{A}$. Given V_1 , position 4 is already $\mathbf{A}$, so $P(V_4 | V_1) = P(\mathbf{G})P(\mathbf{U})P(\mathbf{A}) = \frac{1}{6} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{54} \neq \frac{1}{162}$.
- (d) Independent. V_1 depends only on positions 1–4 and V_5 only on positions 5–8. These sets do not overlap, and the letters are independent.

■

Problem 19: Independence 3: Solving for Probabilities

(slide 111)

Events A and C are independent, B and C are independent, and A and B are mutually exclusive. Also $P(A \cup C) = \frac{2}{3}$, $P(B \cup C) = \frac{3}{4}$, and $P(A \cup B \cup C) = \frac{11}{12}$. Find $P(A)$, $P(B)$, and $P(C)$.

Solution. Write $a = P(A)$, $b = P(B)$, $c = P(C)$. By inclusion–exclusion and independence ($P(A \cap C) = ac$, $P(B \cap C) = bc$), and since $A \cap B = \emptyset$ (so $P(A \cap B) = P(A \cap B \cap C) = 0$):

$$a + c - ac = \frac{2}{3}, \quad (1)$$

$$b + c - bc = \frac{3}{4}, \quad (2)$$

$$a + b + c - ac - bc = \frac{11}{12}. \quad (3)$$

Computing $(1) + (2) - (3)$ gives $c = \frac{2}{3} + \frac{3}{4} - \frac{11}{12} = \frac{1}{2}$. Then (1) gives $a + \frac{1}{2} - \frac{a}{2} = \frac{2}{3}$, so $a = \frac{1}{3}$, and (2) gives $b + \frac{1}{2} - \frac{b}{2} = \frac{3}{4}$, so $b = \frac{1}{2}$. Check (3): $\frac{1}{3} + \frac{1}{2} + \frac{1}{2} - \frac{1}{6} - \frac{1}{4} = \frac{11}{12}$. ✓

$$P(A) = \frac{1}{3}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{2}.$$

■

3.5 Law of Total Probability

Problem 20: LOTP 1: Marbles and Buckets

(slide 114)

Step 1: draw a marble from a bag with 1 red, 2 green, and 1 blue marble. The color picks a bucket.
Step 2: draw a marble from that bucket. The probability of drawing blue is $\frac{1}{4}$ from the red bucket, 0 from the green bucket, and $\frac{1}{3}$ from the blue bucket. Let A be the event that the Step 2 marble is blue.

- (a) Find $P(A)$.
- (b) Find $P(\text{red bucket} | A)$.

Solution. Let R, G, B be the bucket chosen in Step 1: $P(R) = \frac{1}{4}$, $P(G) = \frac{2}{4} = \frac{1}{2}$, $P(B) = \frac{1}{4}$.

(a) By the Law of Total Probability,

$$\begin{aligned} P(A) &= P(A | R)P(R) + P(A | G)P(G) + P(A | B)P(B) \\ &= \frac{1}{4} \cdot \frac{1}{4} + 0 \cdot \frac{1}{2} + \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{16} + \frac{1}{12} = \frac{7}{48} \approx 0.146. \end{aligned}$$

(b) By Bayes' Theorem,

$$P(R | A) = \frac{P(A | R)P(R)}{P(A)} = \frac{1/16}{7/48} = \frac{3}{7} \approx 0.43.$$

Check: $P(B | A) = \frac{1/12}{7/48} = \frac{4}{7}$ and $P(G | A) = 0$, which sum to 1 with $\frac{3}{7}$.

■

Problem 21: LOTP 2: The Late Cat, continued

(slide 117)

Use the setup of Chain Rule 1: $P(R) = \frac{1}{3}$; $P(T | R) = \frac{1}{2}$, $P(T | R^c) = \frac{1}{4}$; $P(L | R \cap T) = \frac{1}{2}$, $P(L | R^c \cap T^c) = \frac{1}{8}$, and $P(L) = \frac{1}{4}$ in the other two cases.

(b) What is the probability that the cat is late?

(c) Given that the cat is late, what is the probability that it rained?

Solution.

(b) Condition on the four rain/traffic cases (Law of Total Probability plus the chain rule):

$$\begin{aligned} P(L \cap R \cap T) &= \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{12}, & P(L \cap R \cap T^c) &= \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{24}, \\ P(L \cap R^c \cap T) &= \frac{2}{3} \cdot \frac{1}{4} \cdot \frac{1}{4} = \frac{1}{24}, & P(L \cap R^c \cap T^c) &= \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{8} = \frac{1}{16}. \end{aligned}$$

So $P(L) = \frac{4}{48} + \frac{2}{48} + \frac{2}{48} + \frac{3}{48} = \frac{11}{48} \approx 0.229$.

(c) $P(R \cap L) = \frac{1}{12} + \frac{1}{24} = \frac{1}{8} = \frac{6}{48}$, so

$$P(R | L) = \frac{P(R \cap L)}{P(L)} = \frac{6/48}{11/48} = \frac{6}{11} \approx 0.545.$$

■

3.6 Bayes' Theorem

Problem 22: Bayes 1: SARS Test

(slide 120)

A test detects SARS 98% of the time in people who have it, and has a 1% false-positive rate in people who do not. 0.5% of the population has SARS.

(a) What is $P(\text{SARS} | \text{positive test})$?

(b) What is $P(\text{SARS} | \text{negative test})$?

Solution. Let F = has SARS and E = positive test: $P(F) = 0.005$, $P(E | F) = 0.98$, $P(E | F^c) = 0.01$.

(a) $P(E) = 0.98(0.005) + 0.01(0.995) = 0.0049 + 0.00995 = 0.01485$, so

$$P(F | E) = \frac{P(E | F)P(F)}{P(E)} = \frac{0.0049}{0.01485} \approx 0.330.$$

(b) $P(E^c) = 0.02(0.005) + 0.99(0.995) = 0.0001 + 0.98505 = 0.98515$, so

$$P(F | E^c) = \frac{P(E^c | F)P(F)}{P(E^c)} = \frac{0.0001}{0.98515} \approx 0.0001.$$

■

Problem 23: Bayes 2: Fairness

(slide 123)

A demographic group D makes up 10% of the population. A health condition H affects 5% of people in D and 1% of people outside D .

- (a) What is $P(H | D)$?
- (b) What is $P(D | H)$?
- (c) A naive model always predicts “no condition.” On a representative sample, how often is it right?

Solution.

(a) Given directly: $P(H | D) = 0.05$.

(b) $P(H) = P(H | D)P(D) + P(H | D^c)P(D^c) = 0.05(0.1) + 0.01(0.9) = 0.005 + 0.009 = 0.014$, so

$$P(D | H) = \frac{P(H | D)P(D)}{P(H)} = \frac{0.005}{0.014} = \frac{5}{14} \approx 0.357.$$

Check with 10,000 people: 50 cases in D and 90 outside, so $\frac{50}{140} = \frac{5}{14}$.

- (c) It is right whenever the person does not have the condition: $P(H^c) = 1 - 0.014 = 0.986$, or 98.6%. Yet it never helps a single person who has the condition, so high accuracy alone is misleading.

■

4 Other Questions in the Deck (not tagged PRACTICE)

4.1 “Why CS 109?” Teasers

Problem 24: Birthday Problem

(slides 11–12; wording reconstructed)

How many people must be in a room before the probability that at least two of them share a birthday exceeds $\frac{1}{2}$? (Assume 365 equally likely birthdays and ignore leap years.)

Solution. With n people, fill in birthdays one at a time. The $(k + 1)$ -st person avoids the first k birthdays with probability $\frac{365-k}{365}$, so

$$P(\text{no shared birthday}) = \prod_{k=0}^{n-1} \frac{365-k}{365}.$$

For $n = 22$ this is about 0.524 (so $P(\text{shared}) \approx 0.476$). For $n = 23$ it is about 0.493 (so $P(\text{shared}) \approx 0.507$). The answer is $n = 23$. ■

Problem 25: Collecting Stickers

(slides 13–14; wording reconstructed)

There are 10 different stickers, and each pack contains one sticker chosen uniformly at random. On average, how many packs must you open to collect all 10?

Solution. Once you have k distinct stickers, each new pack gives a new one with probability $\frac{10-k}{10}$. The expected wait for the next new sticker is therefore $\frac{10}{10-k}$. Summing over $k = 0, \dots, 9$:

$$E[T] = \frac{10}{10} + \frac{10}{9} + \dots + \frac{10}{1} = 10 \sum_{j=1}^{10} \frac{1}{j} \approx 29.29.$$

■

Problem 26: 100 Coin Flips

(slides 15–16; wording reconstructed)

Flip a fair coin 100 times. What is the probability that the number of heads is between 40 and 60?

Solution. Let $X \sim \text{Bin}(100, 0.5)$. Then $E[X] = 50$ and $\text{SD}(X) = \sqrt{100 \cdot 0.5 \cdot 0.5} = 5$. By the Central Limit Theorem, X is approximately normal, so

$$P(40 \leq X \leq 60) \approx P(-2 \leq Z \leq 2) \approx 0.9545.$$

(The exact binomial value, which the continuity correction approximates well, is about 0.9648.) ■

Problem 27: Collecting Words

(slide 19; setup inferred, no solution slide in deck)

Each time you push a button, you see one of n words, chosen uniformly at random and independently of earlier pushes. Let X_i be the push on which you first see the i -th *new* word. For example, the pushes *penguin*, *penguin*, *randomized*, *hello*, *randomized*, *spaceship* give $X_1 = 1$, $X_2 = 3$, $X_3 = 4$, $X_4 = 6$. Find $E[X_1]$, $E[X_2 - X_1]$, $E[X_3 - X_2]$, $E[X_i - X_{i-1}]$, and the expected number of pushes needed to see all n words.

Solution. $E[X_1] = 1$, since the first push always shows a new word. After $i - 1$ distinct words have been seen, each push shows a new word with probability $\frac{n-(i-1)}{n}$, so the wait $X_i - X_{i-1}$ is geometric with

$$E[X_i - X_{i-1}] = \frac{n}{n - i + 1}.$$

In particular $E[X_2 - X_1] = \frac{n}{n-1}$ and $E[X_3 - X_2] = \frac{n}{n-2}$. By linearity of expectation,

$$E[X_n] = \sum_{i=1}^n \frac{n}{n - i + 1} = n \left(1 + \frac{1}{2} + \dots + \frac{1}{n} \right) = nH_n \approx n \ln n.$$

■

4.2 Conditional Probability Warm-up (Lecture)

Problem 28: Two Dice, Two Conditionals

(slides 51–52)

Roll two fair dice.

- (a) Given that the sum is 7, what is the probability that the first die is 2?
- (b) Given that the first die is 2, what is the probability that the sum is 7?

Solution.

- (a) The sum is 7 in 6 equally likely outcomes, and only (2, 5) has first die 2: $P(\text{first} = 2 \mid \text{sum} = 7) = \frac{1}{6}$.
- (b) Given first die 2, the sum is 7 only if the second die is 5: $P(\text{sum} = 7 \mid \text{first} = 2) = \frac{1}{6}$.

Did those answers have to match? Not in general: $P(E \mid F) = P(F \mid E) \frac{P(E)}{P(F)}$. They match here only because $P(\text{sum} = 7) = P(\text{first} = 2) = \frac{1}{6}$. This sets up Bayes' Theorem. ■